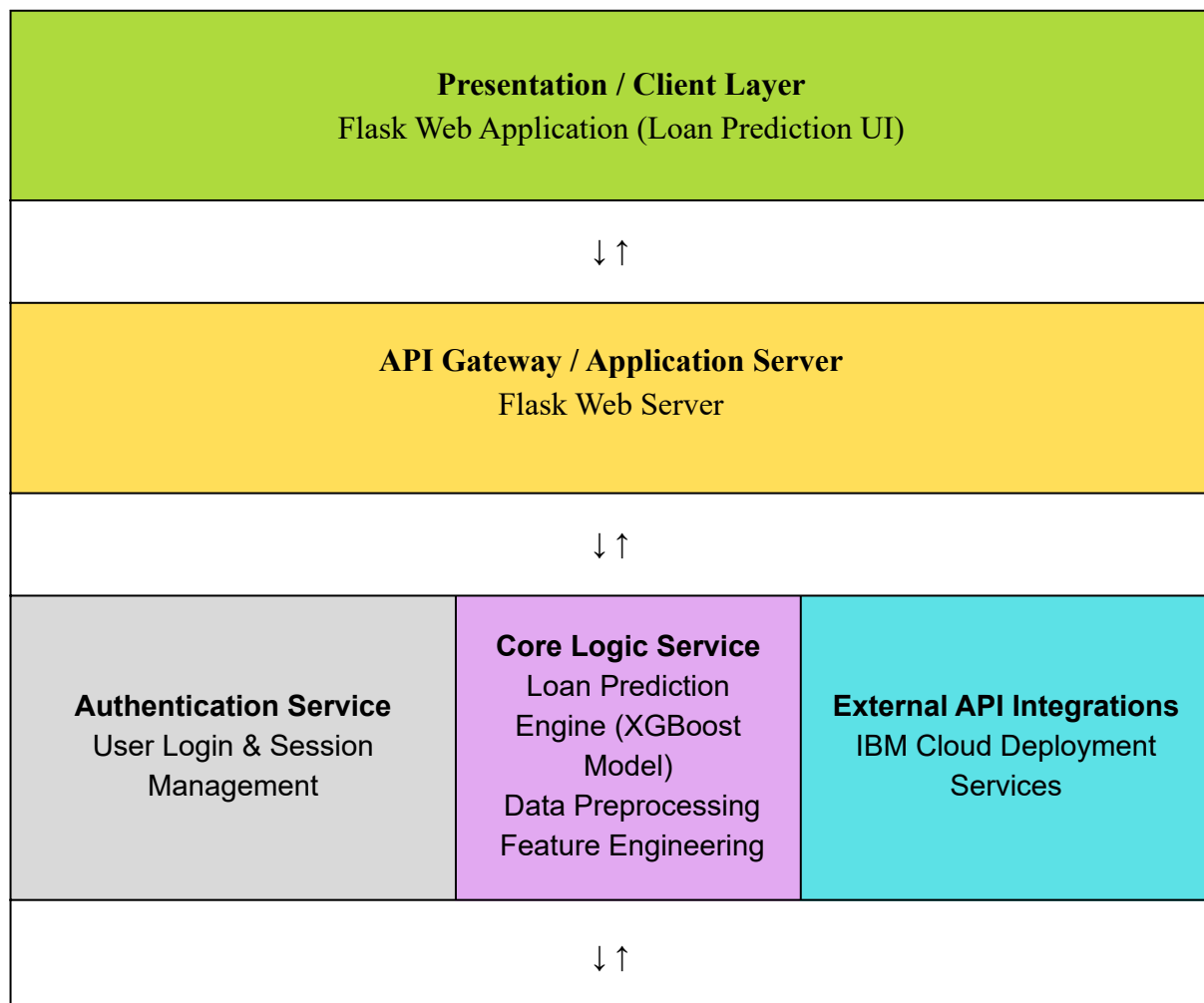


Solution Architecture

Date	27 June 2026
Team ID	-
Project Name	Smart Lender
Maximum Marks	5 Marks

1. Solution Architecture Diagram

The proposed solution architecture represents the interaction between the user interface, application server, machine learning model, and data storage components. It demonstrates how applicant information is processed, analyzed, and classified to predict loan eligibility in real time. This layered architecture ensures scalability, accuracy, and efficient loan decision-making for banking and financial institutions.



Data / Storage Layer Loan Prediction Dataset (CSV) Trained ML Model (.pkl/.jobli)
--

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a web interface for users to enter applicant details and view loan approval predictions.	Flask, HTML, CS
API Gateway	Handles incoming user requests and routes them to the prediction engine.	Flask Web Server
Core Logic Service	Processes applicant data, performs preprocessing, and predicts loan approval using the trained XGBoost model.	Python, Scikit-learn, XGBoost, Pandas, NumPy
External API Integration	Supports deployment and hosting of the application in a cloud environment.	IBM Cloud
Data Layer	Stores the loan dataset and the trained machine learning model used for prediction.	CSV Dataset, Joblib/Pickle